

Homework 4

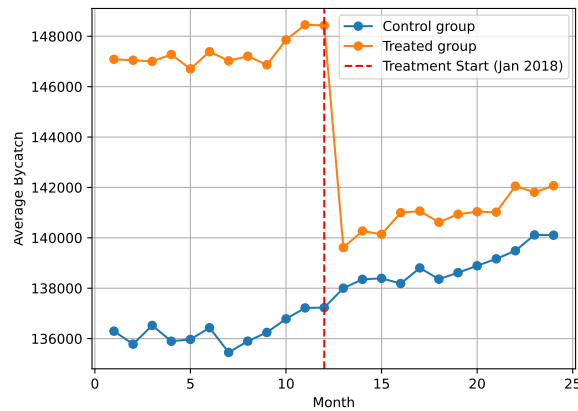
Tannishtha Ghosh

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1 Python

1. Upon visual inspection of Figure 1, it appears there are parallel trends before treatment.

Figure 1: Trends in average bycatch over time for the control and treated groups.



Notes: The figure plots average bycatch levels for the control and treated groups over 24 months from January 2017 to December 2018. The treatment intervention was introduced in January 2018, as indicated by the vertical line. The y-axis represents average bycatch, and the x-axis represents months.

2. By comparing the change in bycatch levels over time for the treatment group and control group, the DID estimator calculates the program's causal effect. Assuming that any pre-existing trends influencing bycatch are shared by both groups according to the parallel trends assumption, it seeks to isolate the causal effect of the treatment by deducting the change in bycatch in the control group from the change in bycatch in the treated group.

DID Estimate

-9591.35

3. Comparing the estimates across the difference-in-difference regression models in Table 1:

- Column (a) focuses only on December 2017 as the pre-period and January 2018 as post. It suggests that, firms in the treatment group reduced their bycatch by about 9,591 units after the program was introduced, compared to the control group.
- Column (b) incorporates all available months and includes monthly fixed effects. The treatment effect is still negative (-8,956.78) and statistically significant, though slightly smaller in magnitude than in (a). This suggests that, even after controlling for time trends with month fixed effects, the program still led to a significant reduction in bycatch.

- Column (c) adds firm size and pounds of shrimp and salmon harvested as controls. Firm Size has a negative but insignificant effect (-2,119.71), suggesting that bycatch levels are not necessarily correlated to firm size. Effects of salmon (0.602) and shrimp (1.055) are both positive and significant implying that bycatch is correlated with the volume of seafood harvested. The treatment effect remains negative (-8,436.28) and highly significant, though slightly reduced compared to (b). This suggests that the program's impact is not driven by changes in firm size or catch composition.

Table 1: Difference-in-Differences Estimates of the Impact of the Program on Bycatch

	(a)	(b)	(c)
Intercept	138001.814*** (18657.796)	136154.047*** (18371.139)	1547.006 (1105.822)
pre	-773.216 (598.685)		
treated	11202.043 (23502.902)	11052.450 (23162.972)	-21.902 (308.244)
treat_post	-9591.350*** (3231.787)	-8956.784*** (3166.921)	-8436.282*** (2823.973)
firmsize			-2119.713 (3406.960)
shrimp			1.055*** (0.053)
salmon			0.602*** (0.210)
R-squared	0.003	0.003	0.991
R-squared Adj.	-0.028	-0.018	0.991
Observations	100	1200	1200
Standard errors in parentheses * p<.1, ** p<.05, ***p<.01			

Notes: This table reports difference-in-differences (DiD) estimates of the impact of the program on bycatch. Column (a) estimates a two-period DiD using December 2017 and January 2018. Column (b) uses the full monthly sample with monthly fixed effects. Column (c) controls for firm size and seafood catch composition. Standard errors are clustered at the firm level.

2 STATA

Table 2: Effect of Treatment on Bycatch Reduction

	(1)	(2)
VARIABLES	OLS Fixed Effects	Within-Transformation
Treatment x Post	-8,085*** (2,619)	
Shrimp	1.552*** (0.178)	
Salmon	-0.680 (1.125)	
Treatment x Post (mean)		-8,085*** (2,564)
Shrimp (mean)		1.552*** (0.175)
Salmon (mean)		-0.680 (1.101)
Observations	1,200	1,200
R-squared	0.996	0.428
Standard errors in parentheses		
* p<.1, ** p<.05, ***p<.01		

Notes: This table reports the difference-in-differences (DiD) estimates of the impact of the program on bycatch. Column (1) estimates estimates the DiD regression with firm and month-specific fixed effects. Column (2) performs the DiD estimation through "within-transformation" of all of the dependent and independent variables by demeaning each variable. Standard errors are clustered at the firm level.

In Table 2, the estimates are same across the regression specifications with minor differences in standard errors.